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C A P S T O N E

ACADEMIC SUCCESS AND DROPOUT EXPLORATION

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Introduction

Essential Question

How can an institution identify students who may need support early enough to improve student outcomes?



Project Objective

Predict college dropout rates given a student's academic and socioeconomic background.



Motivation

Education is crucial to society, which makes predicting dropout rates all the more important. The ability to predict these dropout rates allows us to identify students at risk early, which protects both the student's well-being and the university's own funding and enrollment (Delogu, 2024).



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Data Overview

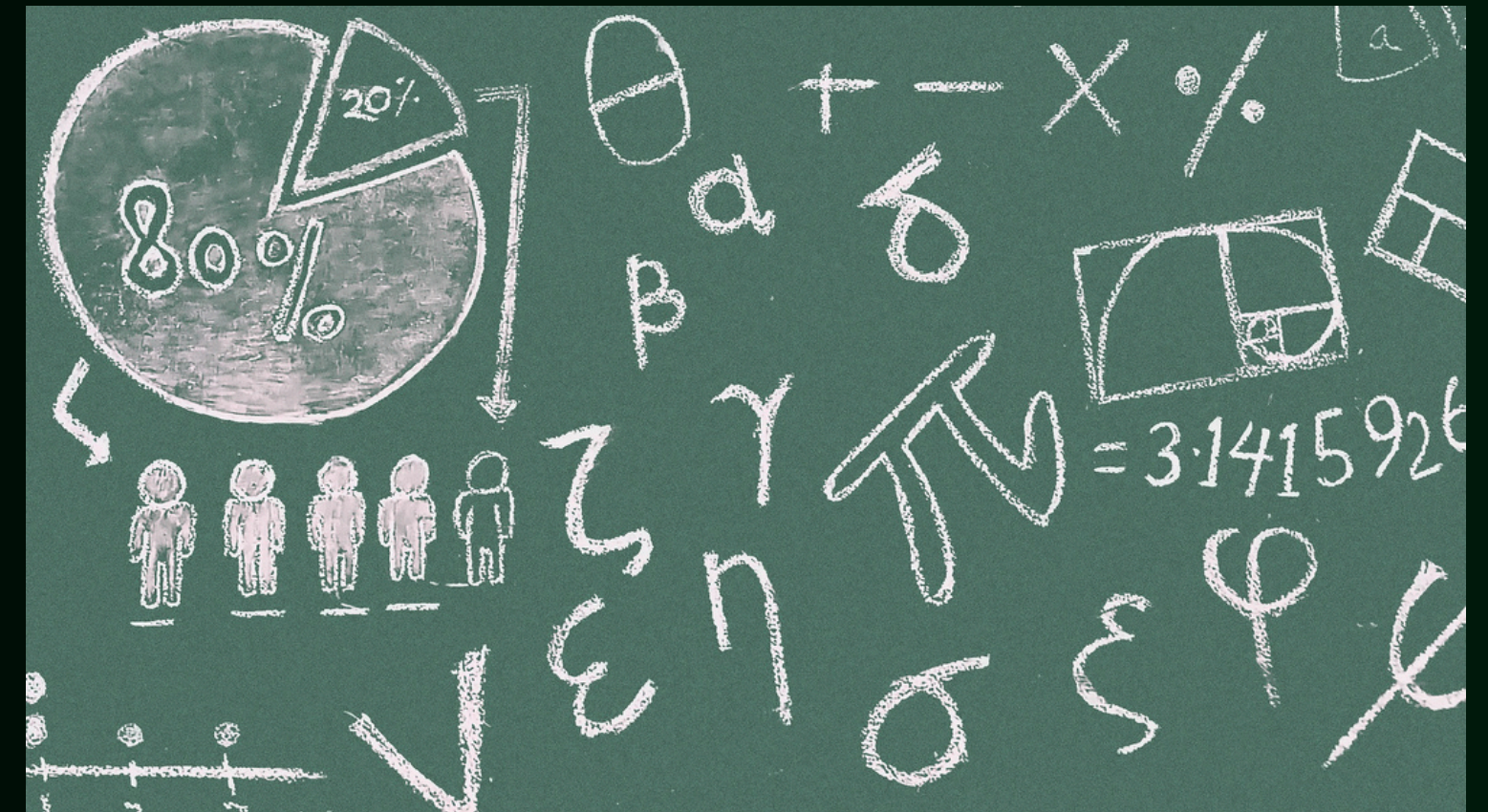
Data Overview

- **36 predictors**
 - **All numerical**
- **4424 rows**
 - **No null or duplicate values**
- **Target variable: Dropout rate**
 - **“Dropout” or “Graduate”**

0	Marital Status	17	Gender
1	Application mode	18	Scholarship holder
2	Application order	19	Age at enrollment
3	Course	20	International
4	Daytime/evening attendance	21	Curricular units 1st sem (credited)
5	Previous qualification	22	Curricular units 1st sem (enrolled)
6	Previous qualification (grade)	23	Curricular units 1st sem (evaluations)
7	Nacionality	24	Curricular units 1st sem (approved)
8	Mother's qualification	25	Curricular units 1st sem (grade)
9	Father's qualification	26	Curricular units 1st sem (without evaluations)
10	Mother's occupation	27	Curricular units 2nd sem (credited)
11	Father's occupation	28	Curricular units 2nd sem (enrolled)
12	Admission grade	29	Curricular units 2nd sem (evaluations)
13	Displaced	30	Curricular units 2nd sem (approved)
14	Educational special needs	31	Curricular units 2nd sem (grade)
15	Debtor	32	Curricular units 2nd sem (without evaluations)
16	Tuition fees up to date	33	Unemployment rate
		34	Inflation rate
		35	GDP

Feature Engineering

- **Approval (passing) rate by semester**
- **Overall approval rate**
- **Grade trend**
- **Financial risk**
- **Scholarships**
 - **Scholarship with no debt**
 - **Scholarship with debt**
- **Age (older students)**
- **One-hot encoding some of the categorical features expressed as numbers with no ordinal meaning**





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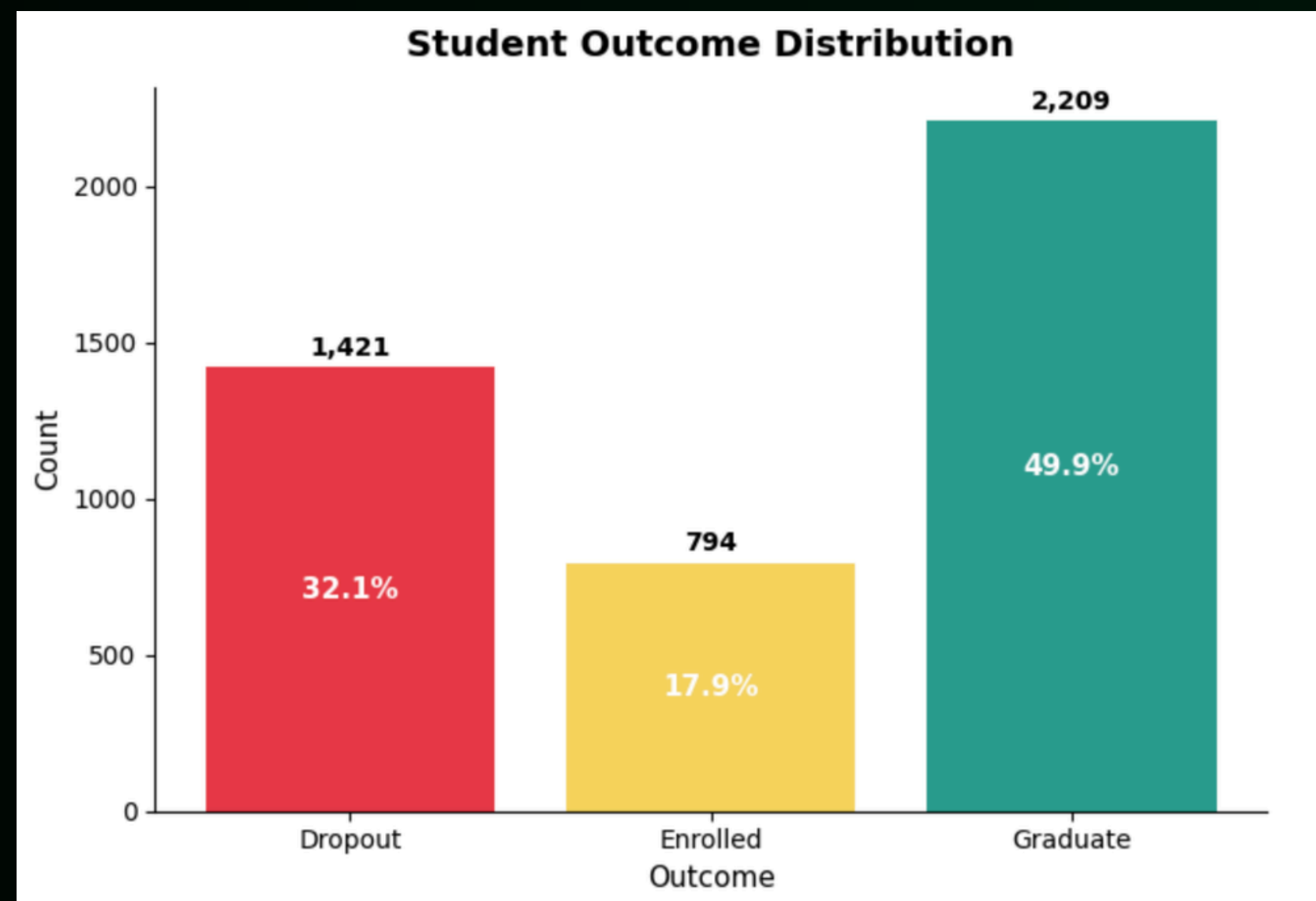
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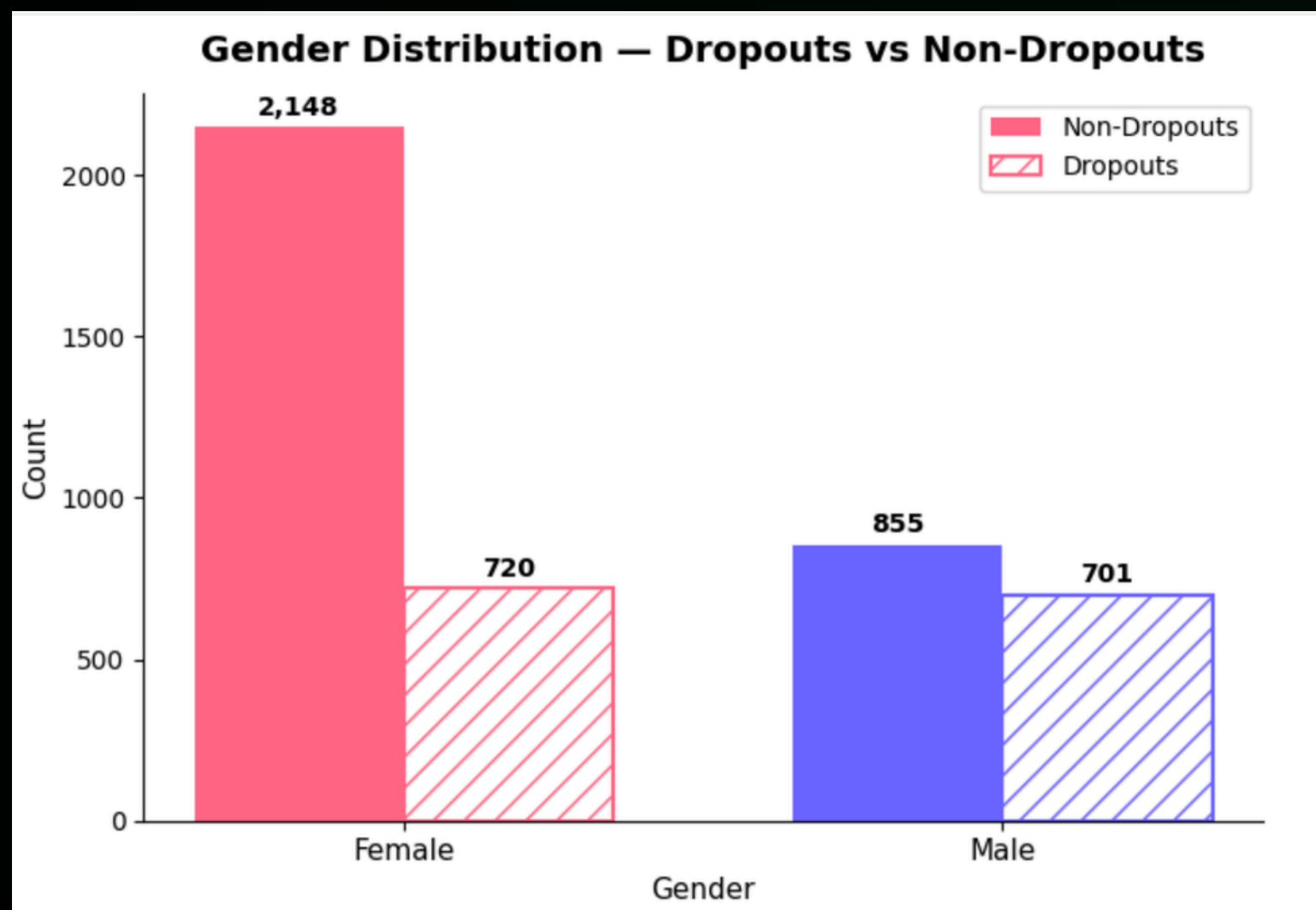
Exploratory Data Analysis

Understanding the Demographics of the Dataset

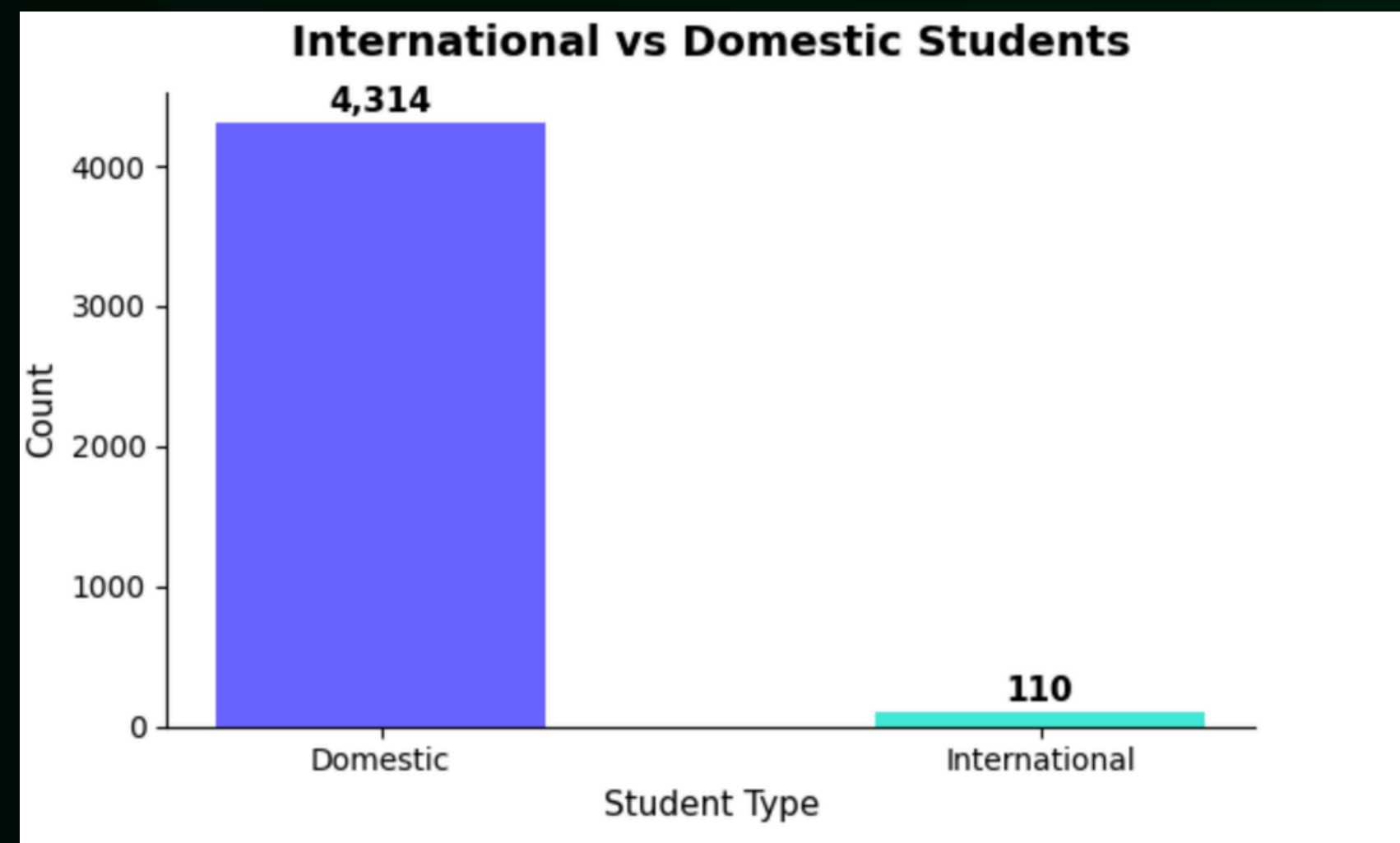
How many students in this dataset dropped out, are currently enrolled, or graduated?



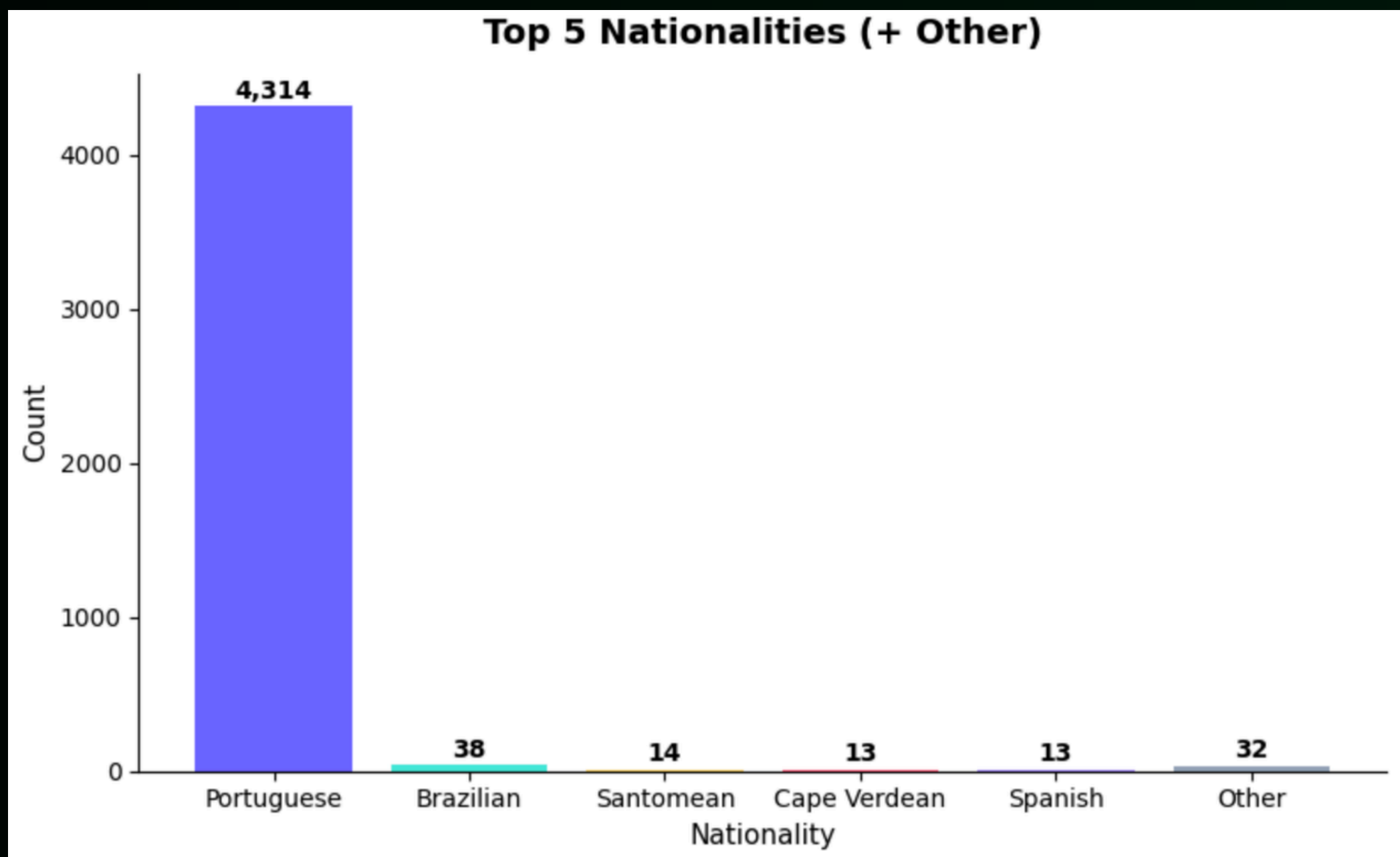
Gender



Origin



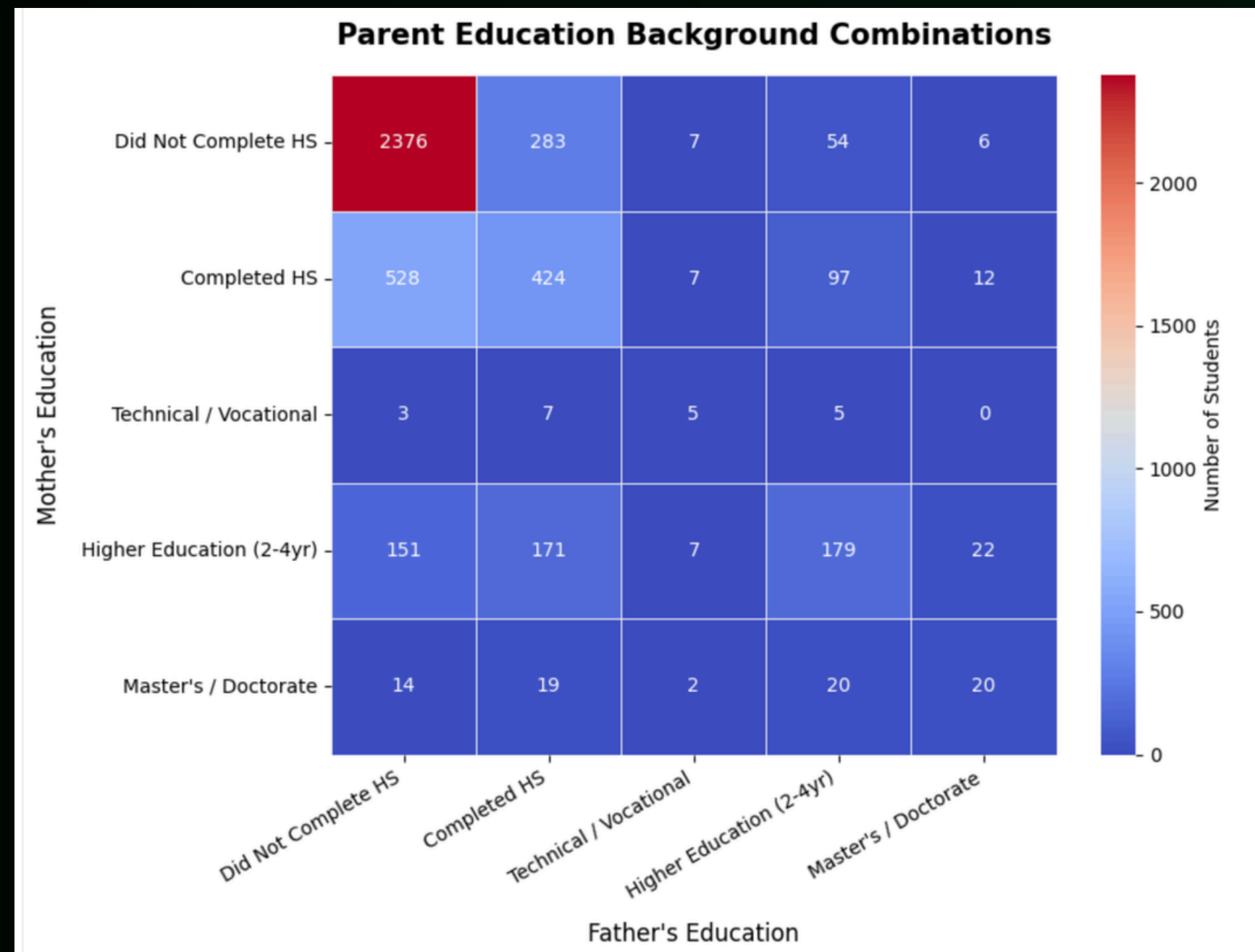
Nationalities of the Students



Looking at Parental Backgrounds of Students

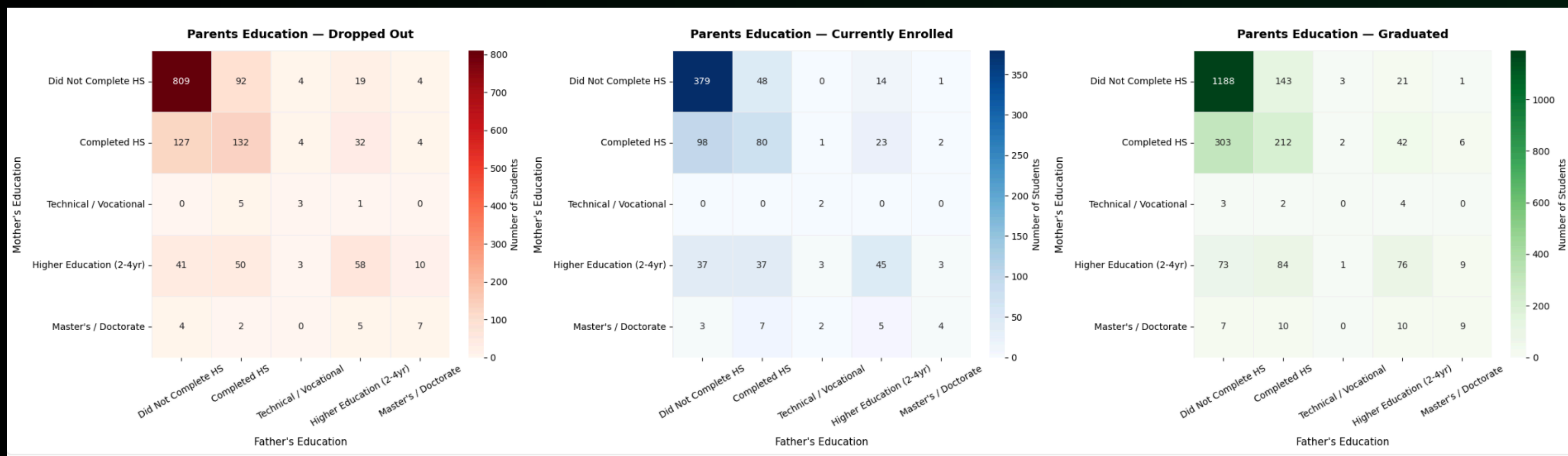
Hypothesis: I hypothesized that parental educational attainment might influence a student's college experience and, as a result, could be associated with whether students ultimately graduated, remained enrolled, or dropped out.

Parent Backgrounds



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Parent Backgrounds

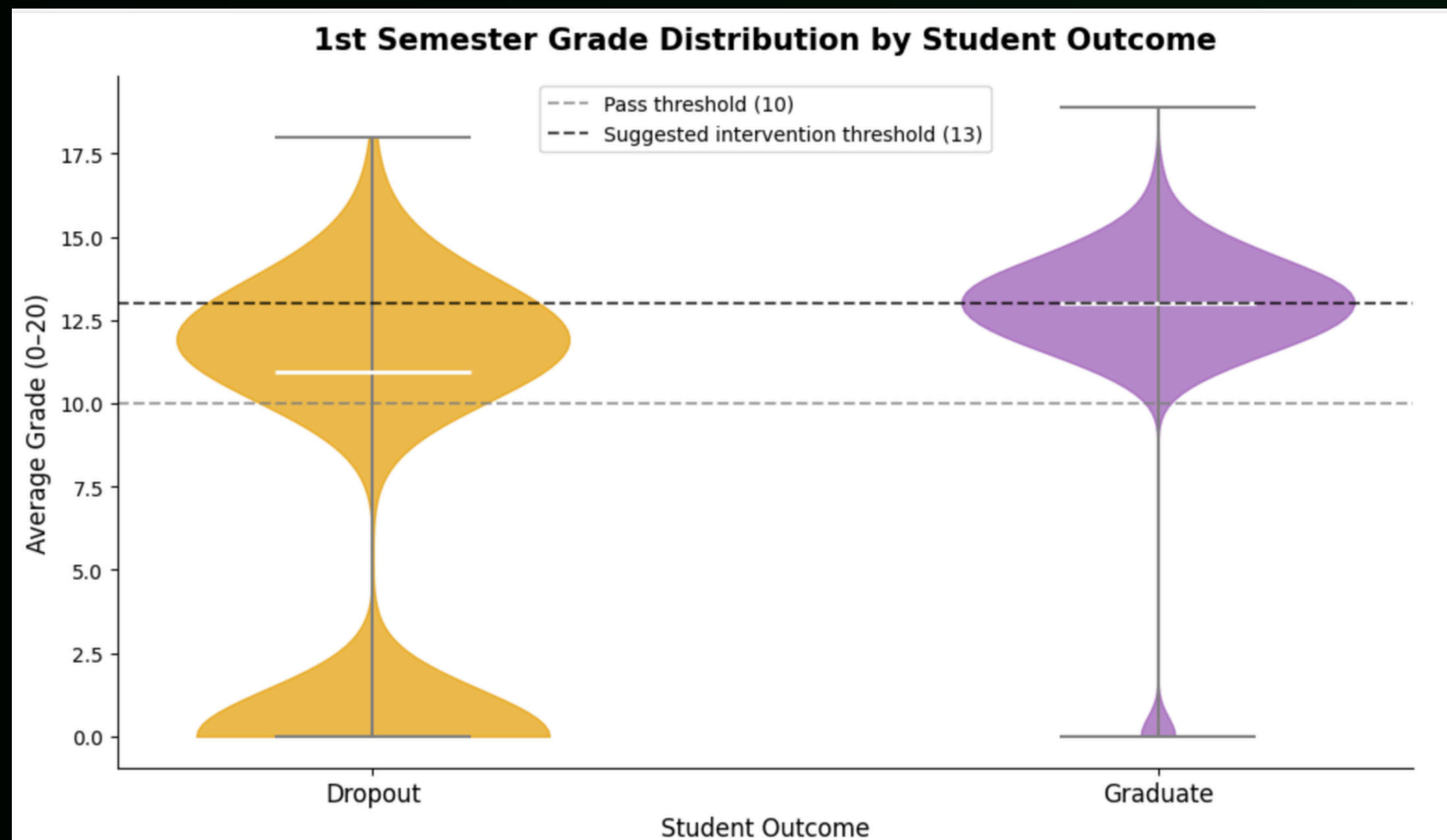


Academic Success and Struggle

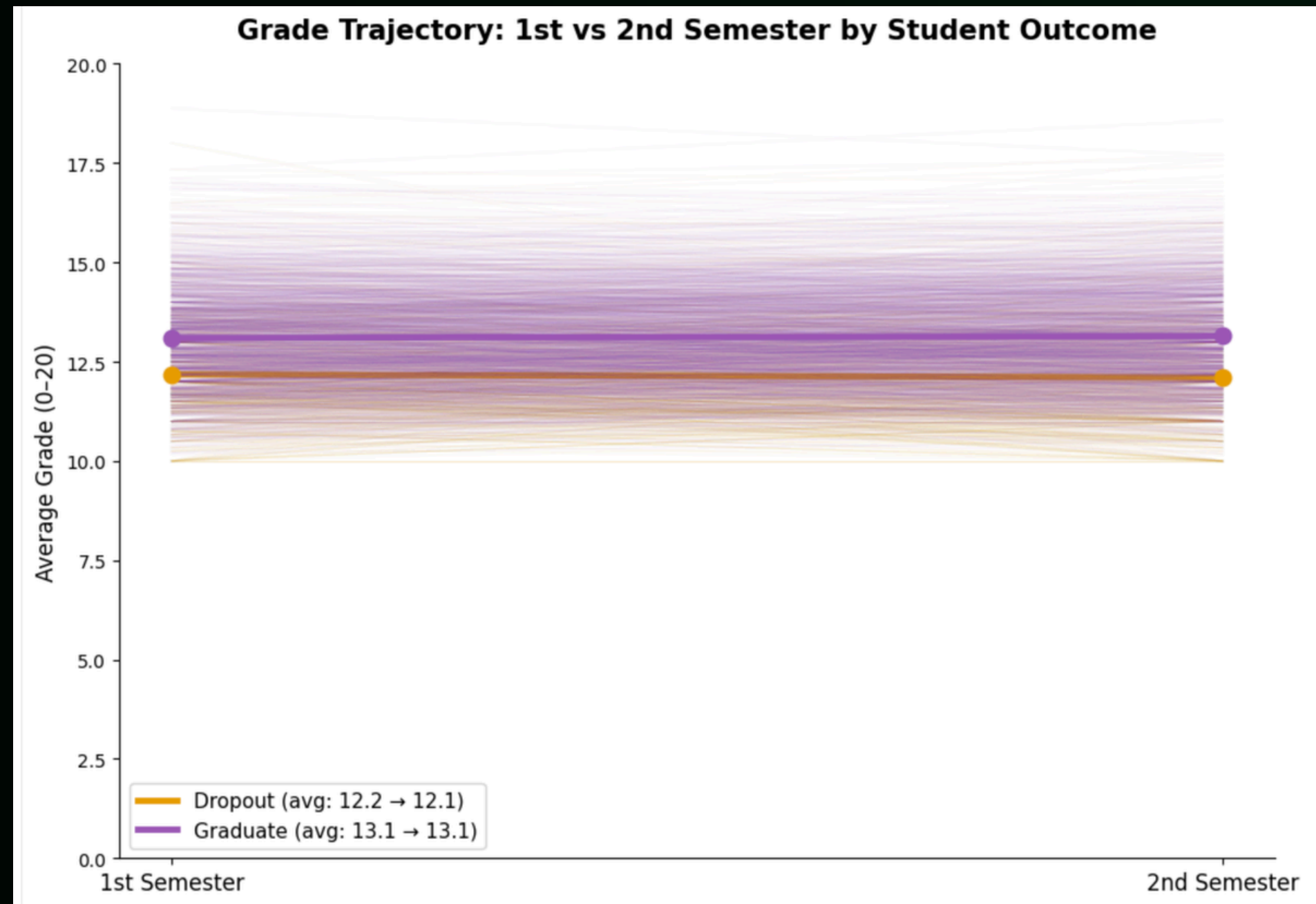
Hypothesis: One of the most intuitive hypotheses was that academic performance would be related to student retention, so I explored whether students who eventually dropped out showed signs of academic struggle during their first semester.

Looking at Academic Struggle

★ note: the white line indicates the median average grade



What about 2nd Semester?



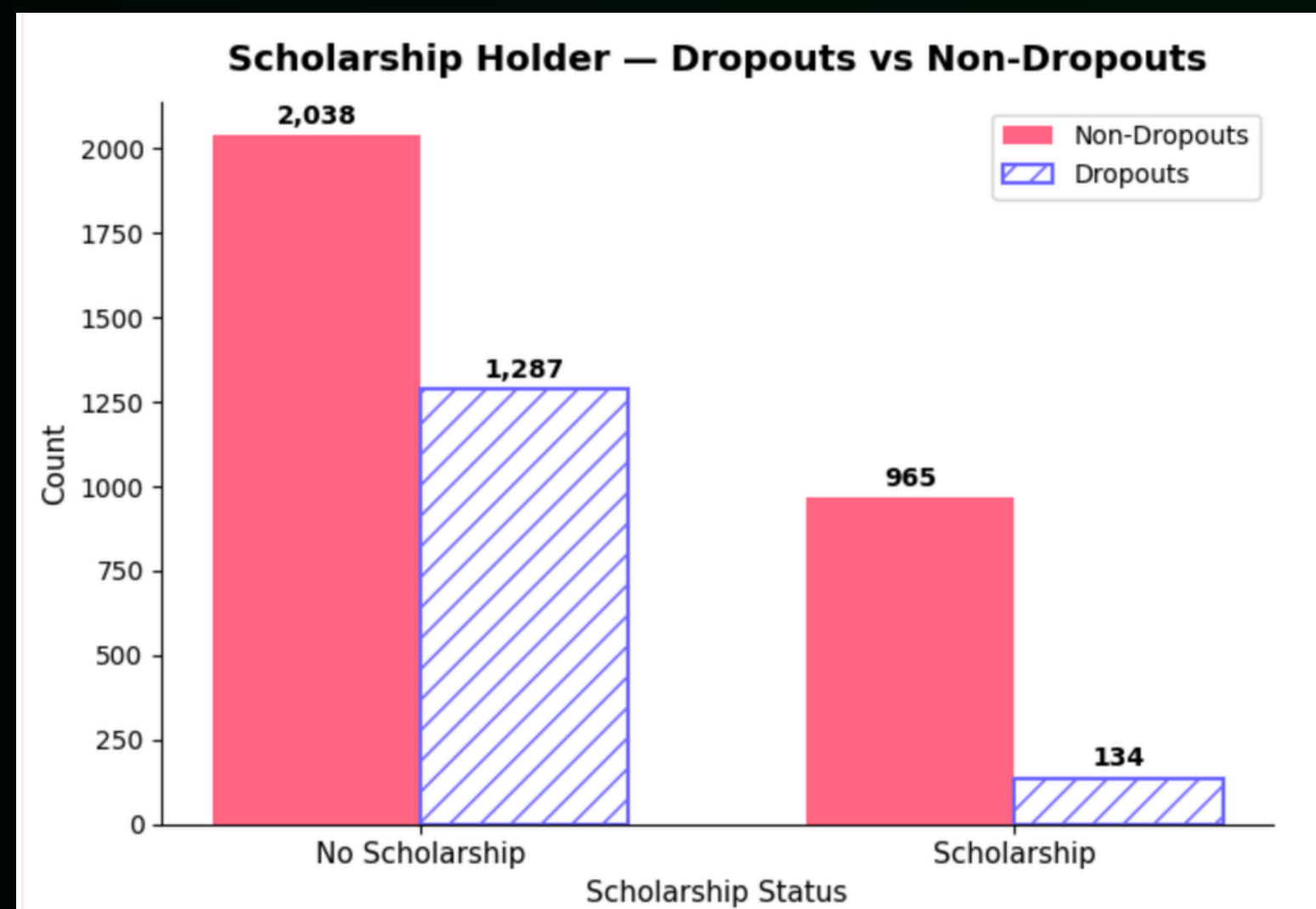
Recommendation #1

Implement an early intervention system for students showing signs of academic struggle.

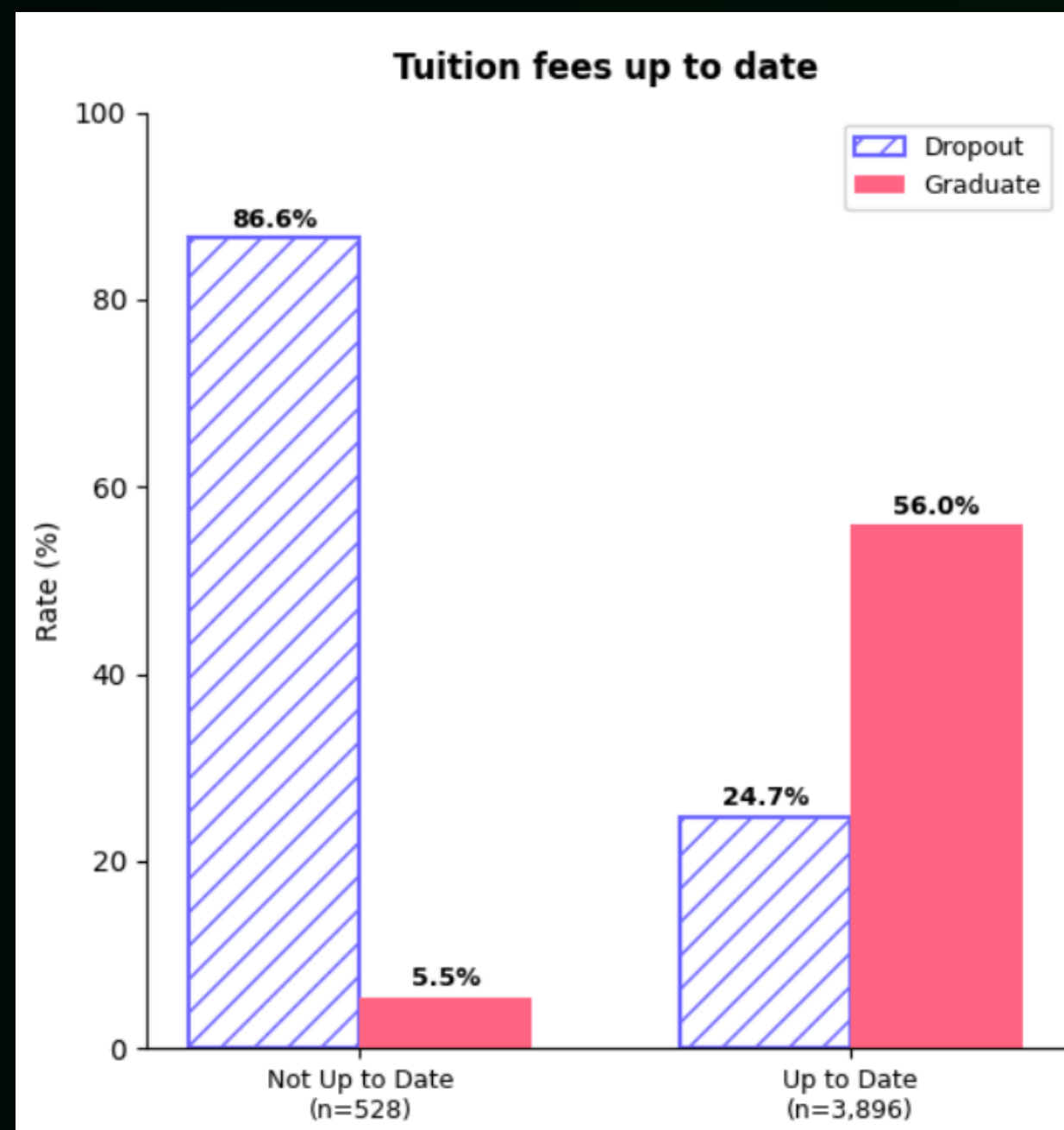
Finances

Hypothesis: Since financial barriers are often cited as a reason students leave higher education, I explored whether students receiving scholarships were more likely to persist and ultimately graduate. Another factor I explored was whether students who were behind on tuition payments were more likely to leave the university than students whose accounts were in good standing.

Scholarship Holders



Finances as an Indicator



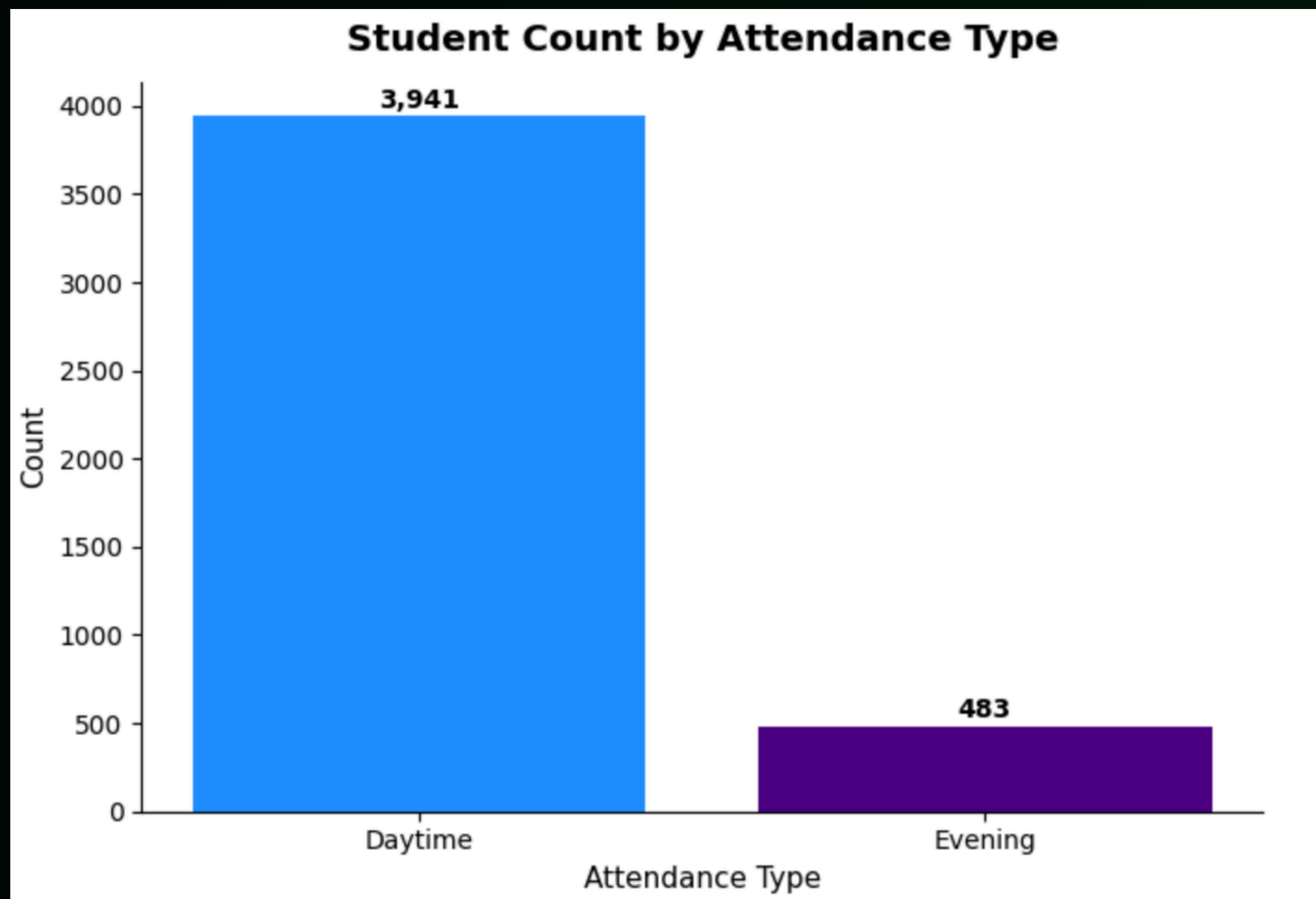
Recommendation #2

Explore expanded financial support opportunities for students at risk of dropping out.

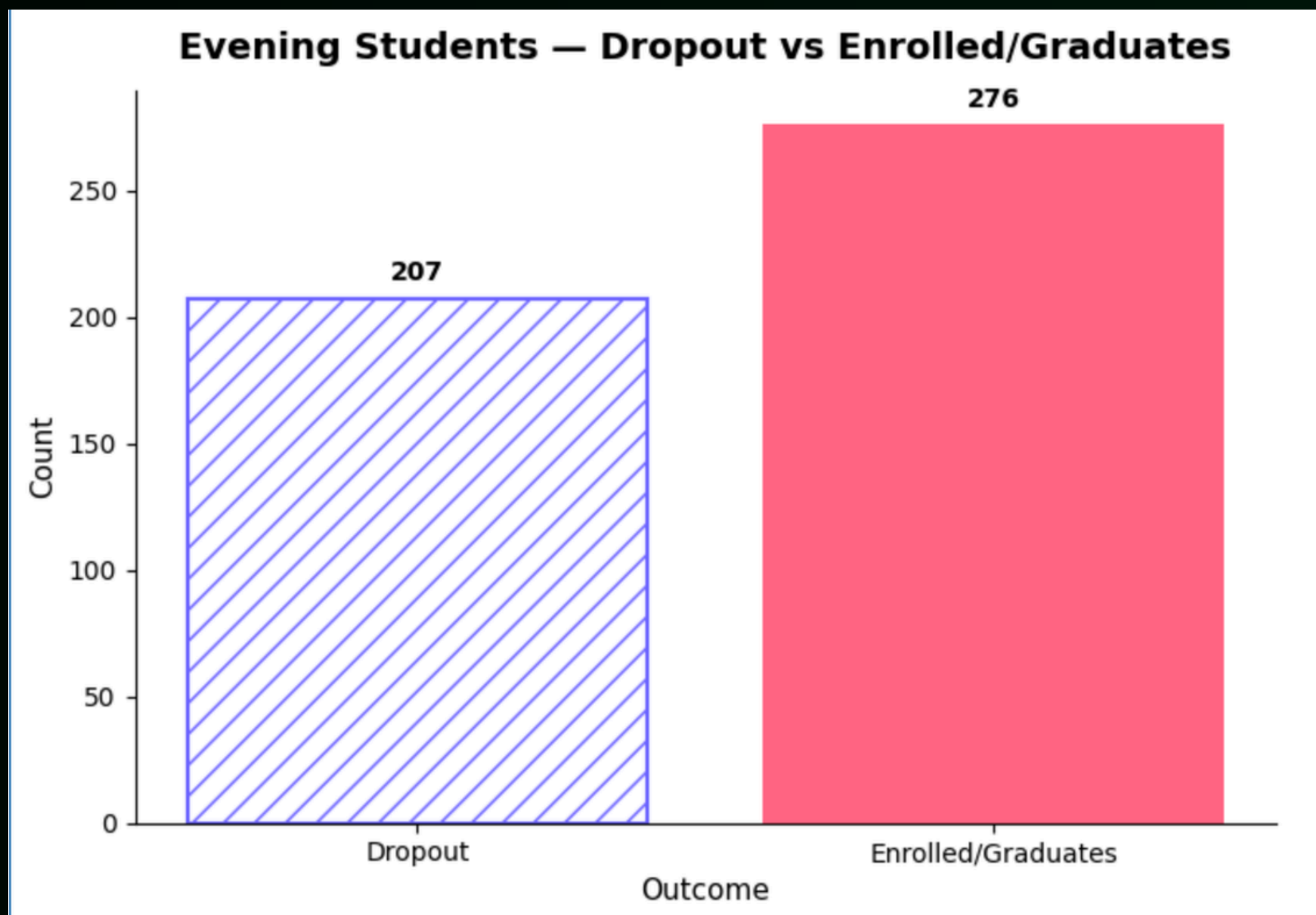
Evening Students

Hypothesis: I hypothesized that enrollment in evening courses might serve as an indicator of different life circumstances and therefore could be associated with student retention outcomes.

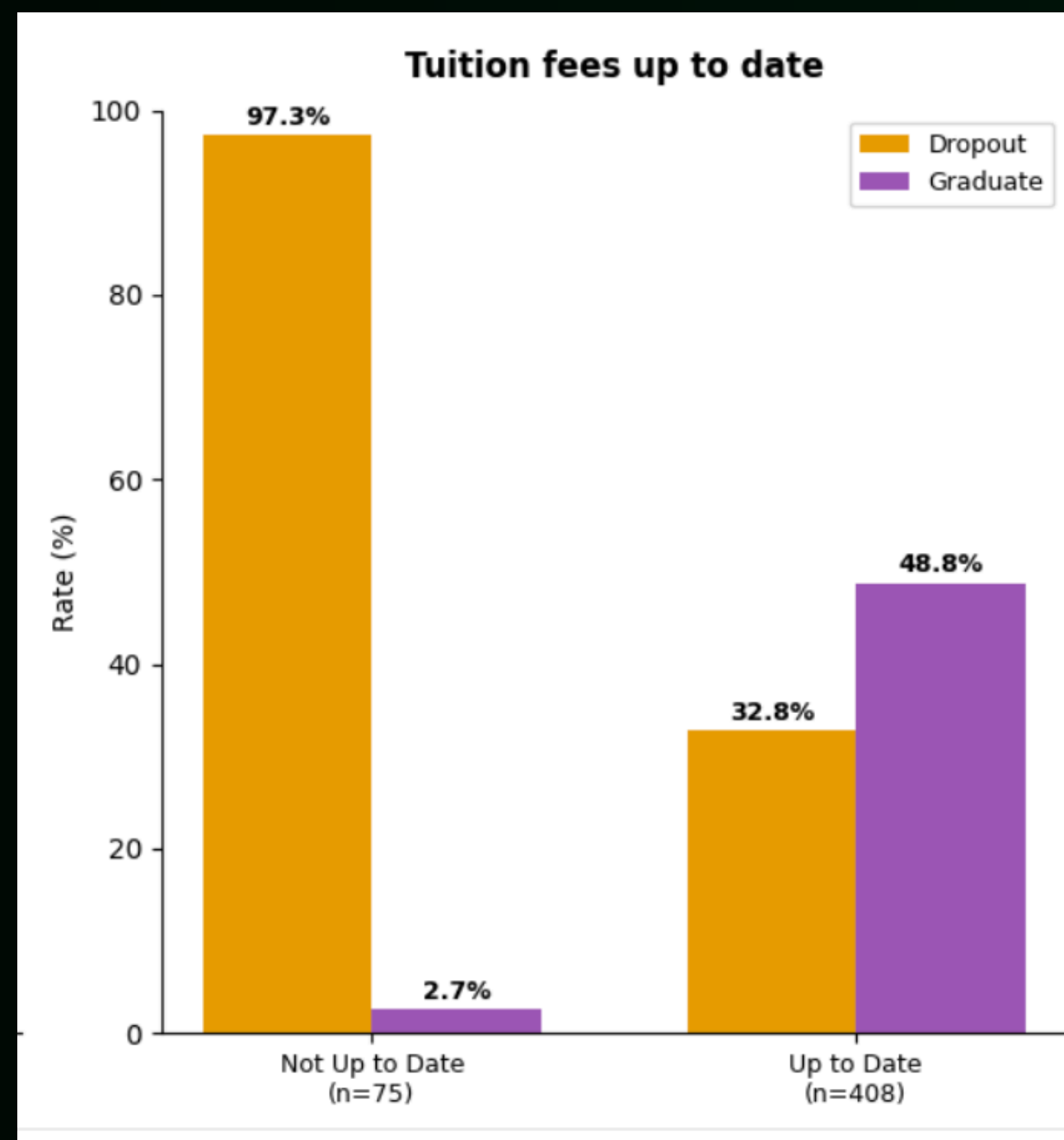
Taking a Closer Look at Evening Students



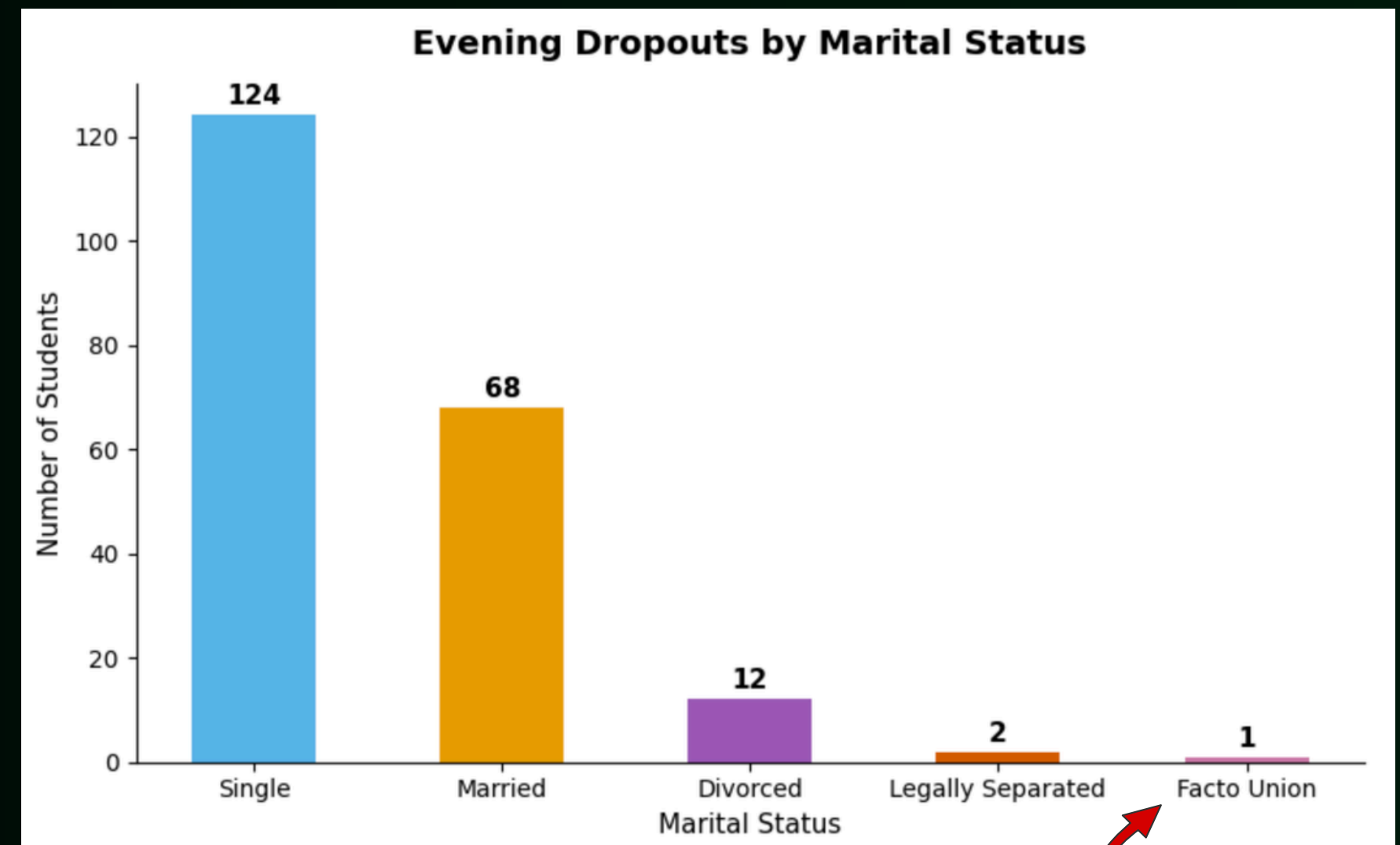
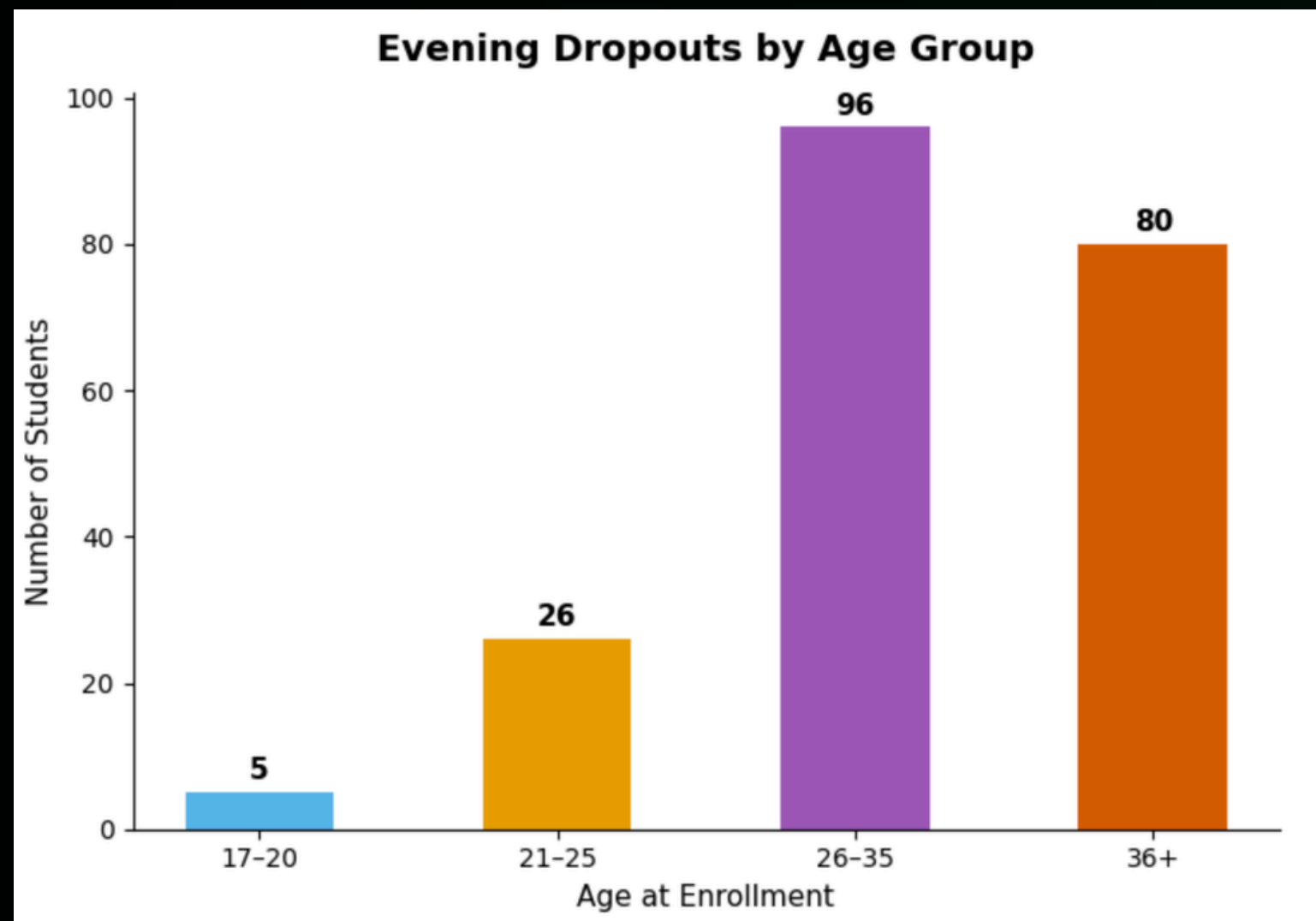
Taking a Closer Look at Evening Students



Tuition Fees ~ Evening Students



Taking a Closer Look at Evening Students



facto union is a legal status where two people live together as a couple and share a household but are not legally married

Recommendation #3

Develop targeted support programs for evening and non-traditional students.



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Our Predictive Model

Model Performance: Logistic Regression vs. Random Forest

76.9% for Logistic Regression, 77.9% for Random Forest. The real story is in how each model does per outcome, not the overall number. Both models are strong at identifying Graduates (86% and 85% f1-score) and decent at Dropouts (78% and 80% f1-score), but both struggle badly with the "Enrolled" category (only 41% and 46% f1-score). "Enrolled" is the hardest class to predict because it's a moving target — an enrolled student hasn't resolved into success or failure yet, so there's less of a clear signal in the data pointing one way or the other.

=== Logistic Regression ===

Accuracy: 0.769

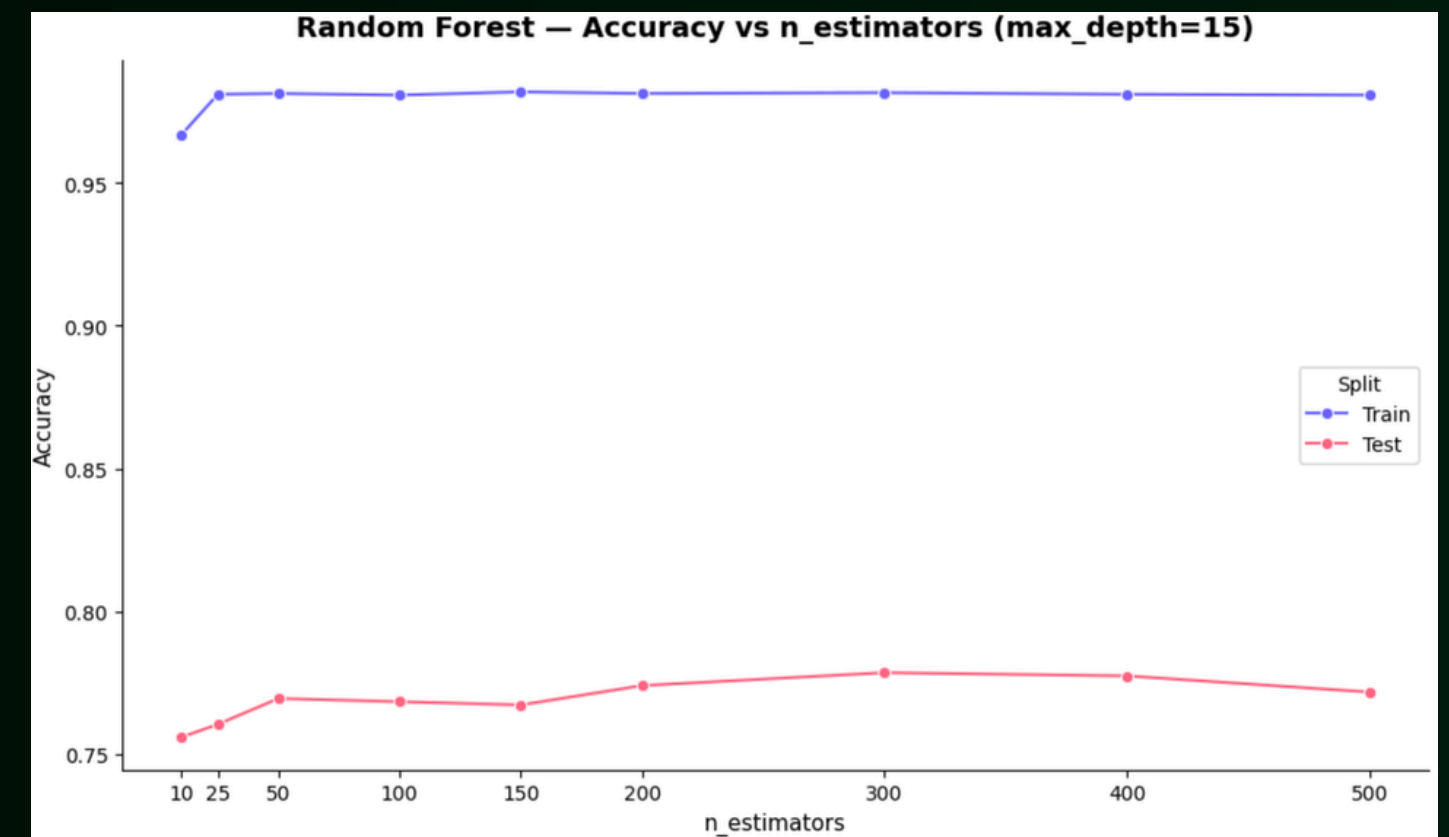
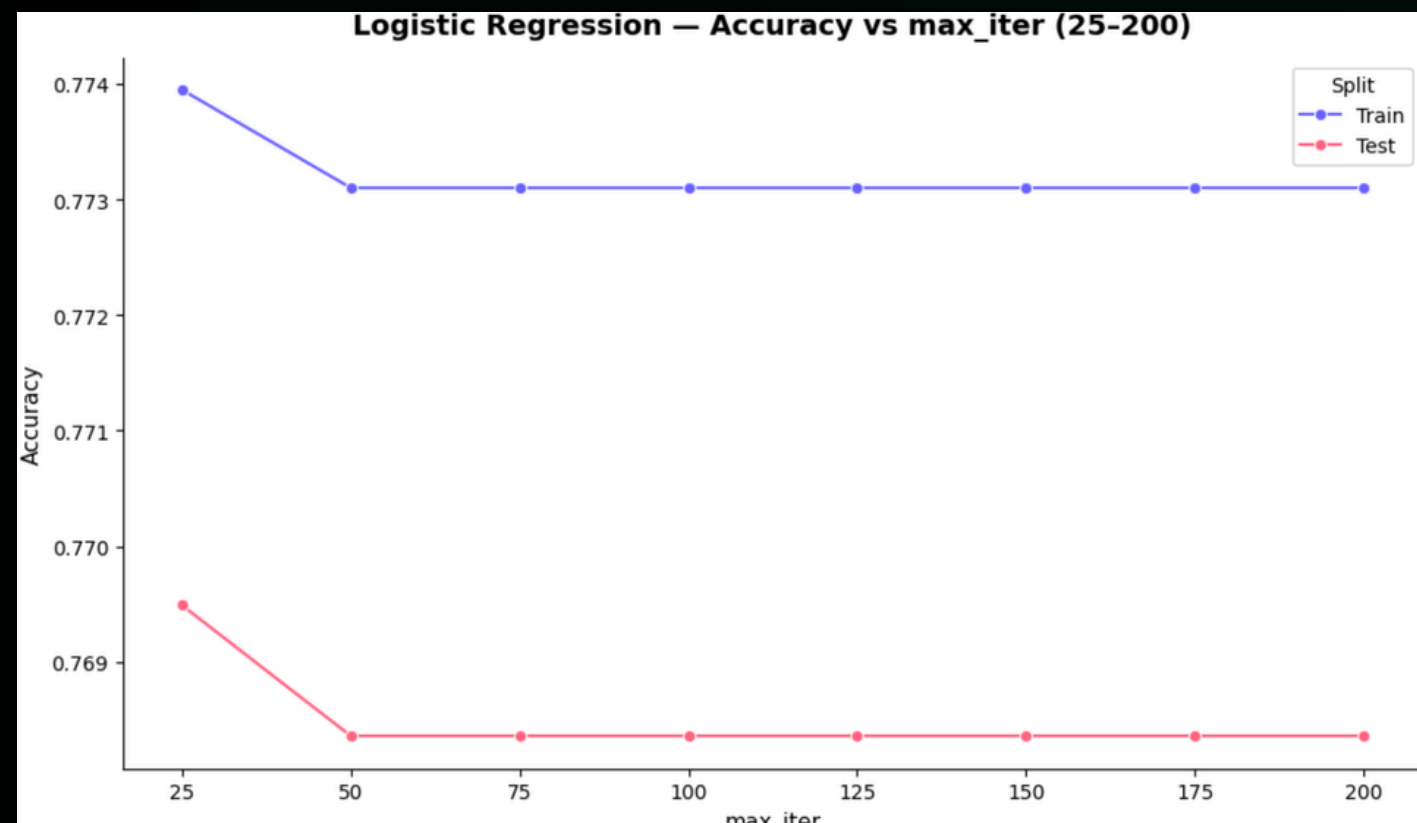
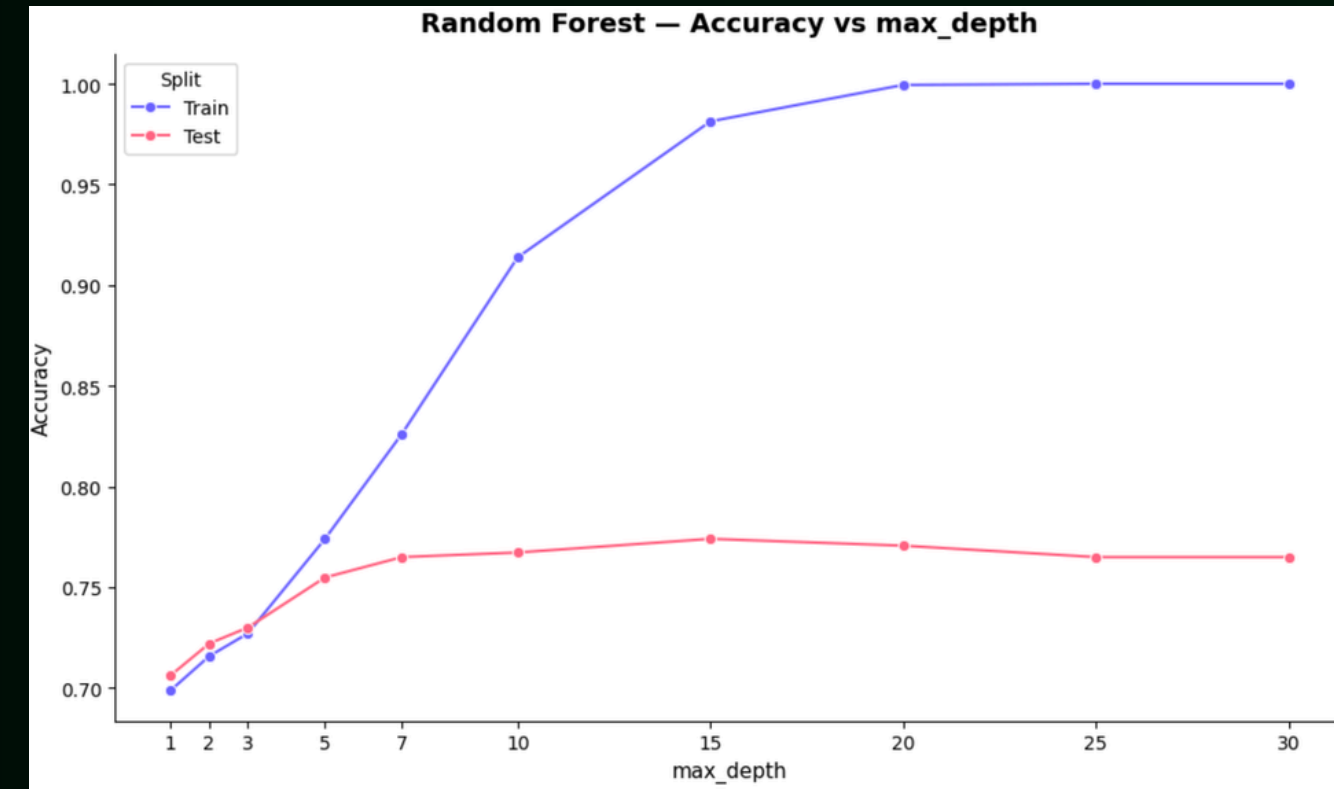
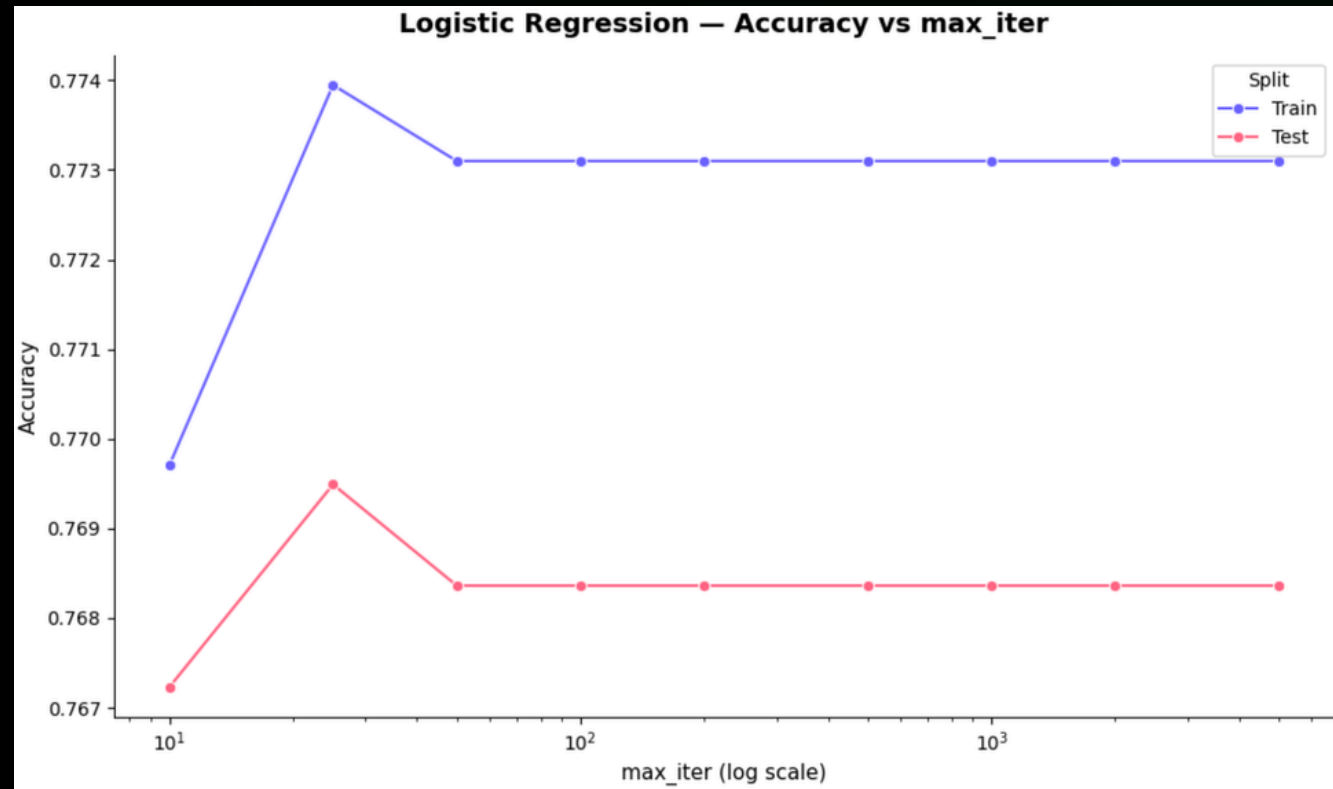
	precision	recall	f1-score	support
Dropout	0.79	0.77	0.78	284
Enrolled	0.54	0.33	0.41	159
Graduate	0.80	0.93	0.86	442
accuracy			0.77	885
macro avg	0.71	0.68	0.68	885
weighted avg	0.75	0.77	0.75	885

=== Random Forest ===

Accuracy: 0.779

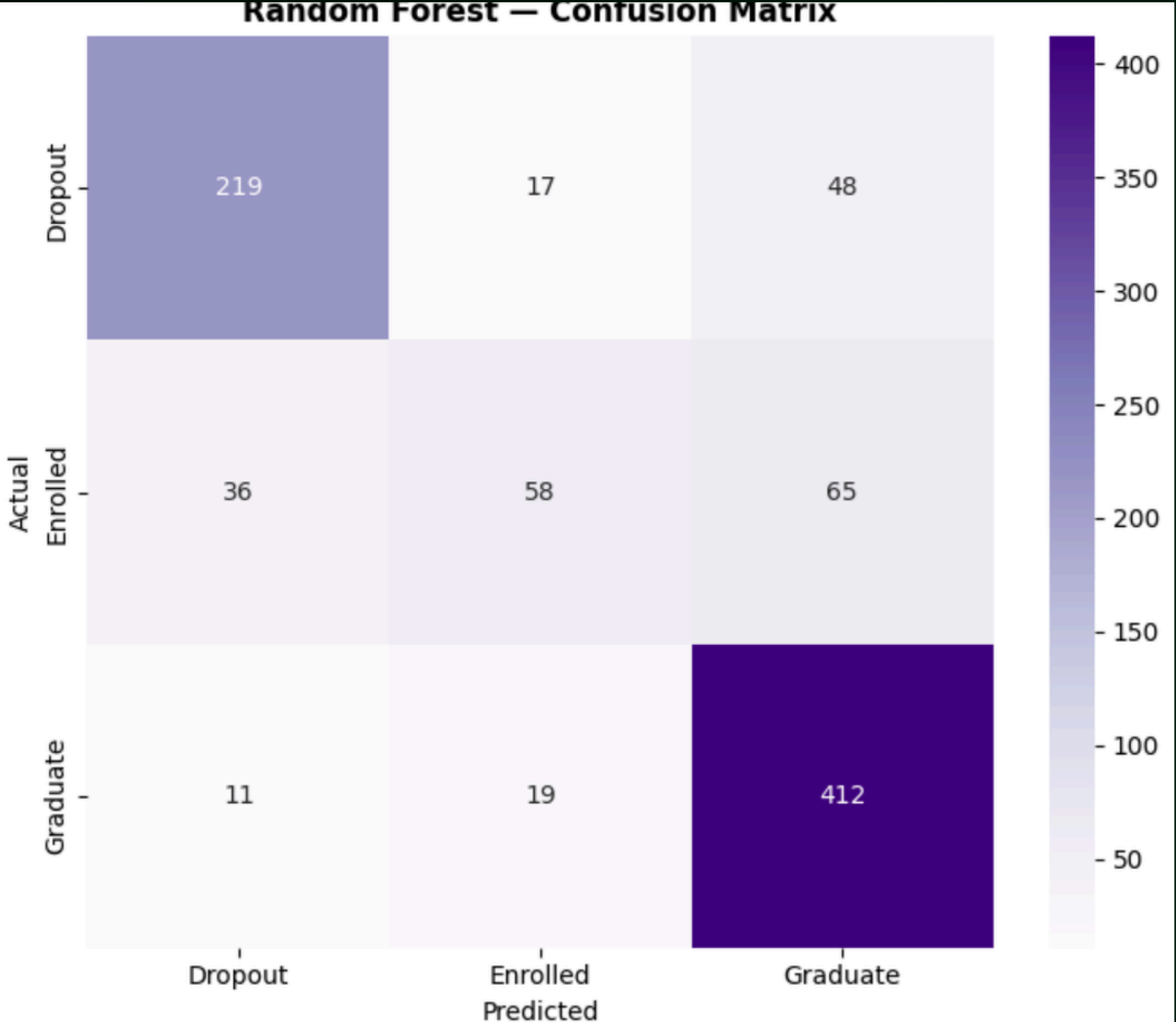
	precision	recall	f1-score	support
Dropout	0.82	0.77	0.80	284
Enrolled	0.62	0.36	0.46	159
Graduate	0.78	0.93	0.85	442
accuracy			0.78	885
macro avg	0.74	0.69	0.70	885
weighted avg	0.77	0.78	0.76	885

Hyperparameter Tuning and Model Optimization



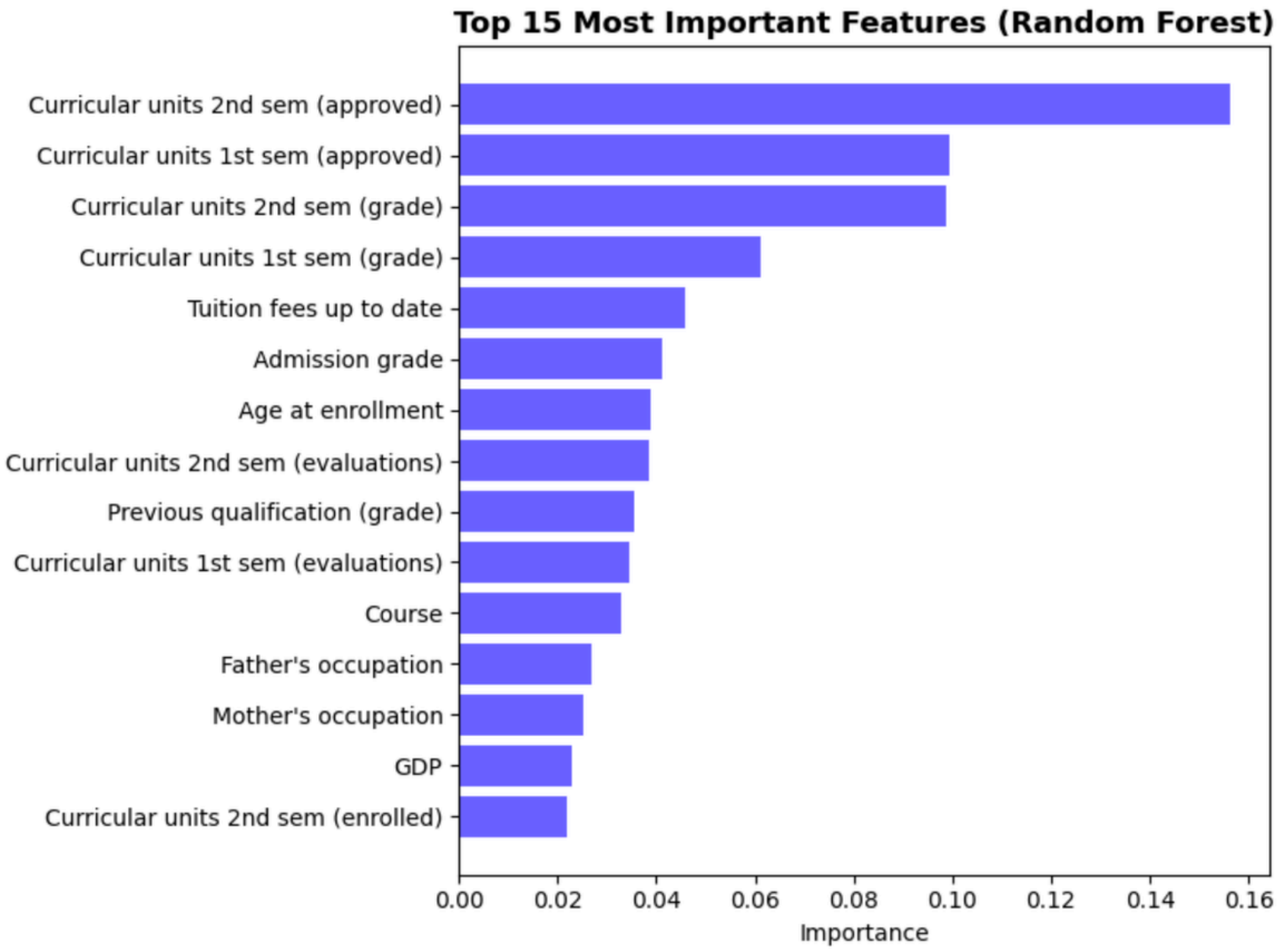
Confusion Matrix: Where Random Forest Gets It Right (and Wrong)

of 284 actual Dropouts, the model correctly caught 219 (75%), but misclassified 48 as Graduates. Of 442 actual Graduates, it correctly identified 412 (94%) – the model's strongest performance by far. Of 159 actual Enrolled students, it only got 58 right (31%), with most of the rest split between being mistaken for Dropouts (36) or Graduates (65). The takeaway: the model is genuinely useful at flagging at-risk dropouts and confidently identifying students on track to graduate, but it can't reliably tell where a currently-enrolled student is headed yet – which actually lines up with real life, since their fate hasn't played out yet.



What Drives the Predictions:
Feature Importance

the model confirms what we found earlier." The top features are almost entirely academic performance metrics — 2nd semester approved units, 1st semester approved units, and grades from both semesters dominate the chart, with tuition payment status (Tuition fees up to date) right behind them. Our model independently rediscovered the same patterns we found by hand in our EDA — early academic performance and financial standing are the strongest predictors of student outcome.





Next Steps



Incorporating a “Risk Index”: Building a risk score that combines multiple factors (e.g. GPA, financial aid status, etc.) into one index and flagging any score that goes below a certain threshold.



Longitudinal Validation: Testing our model’s across different institution types (e.g. community college vs. four-year institution, public vs. private, etc.) to see how our predictions hold up across schools.

Works Cited

Delogu, M., Lagravinese, R., Paolini, D., & Resce, G. (2024). Predicting dropout from higher education: Evidence from Italy. *Economic Modelling*, 130, 106583. <https://doi.org/10.1016/j.econmod.2023.106583>



THANK YOU